Finding 3 - Explore Linux Auditing with auditd (Ubuntu VM)

Objective

Install auditd on Ubuntu, run a command, and locate it in the audit log with `ausearch`.

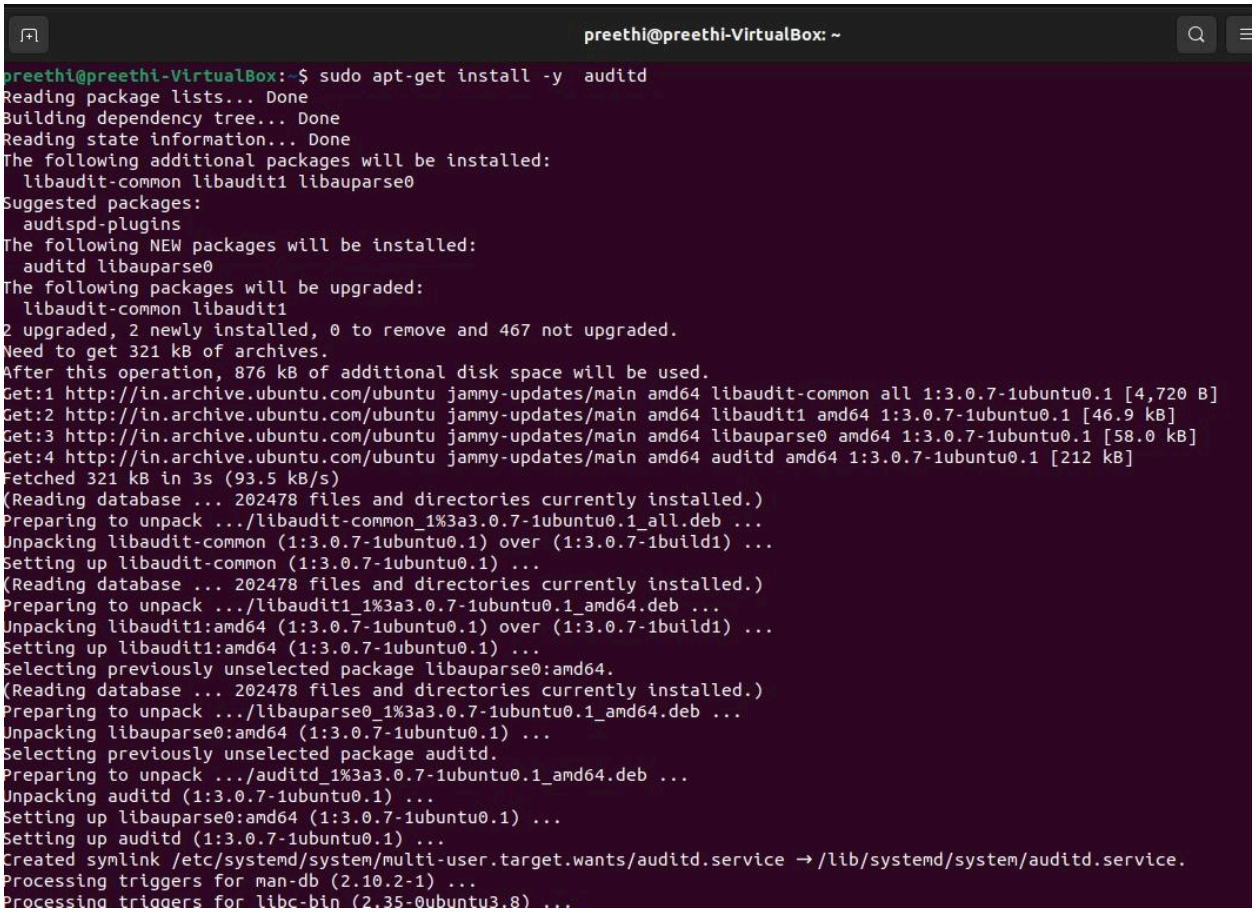
Steps performed

Step	Action	Command
1	Install auditd	`sudo apt-get install -y auditd`
2	Start and enable the service	`sudo systemctl enable auditd --now`
3	Verify the service is running	`sudo systemctl status auditd`
4	Run a command to audit	`sudo apt install nano`
5	Search the audit log	`sudo ausearch -k nano` or `sudo ausearch -m EXECVE -ts recent`
6	Screenshot the matching entry	See screenshots below

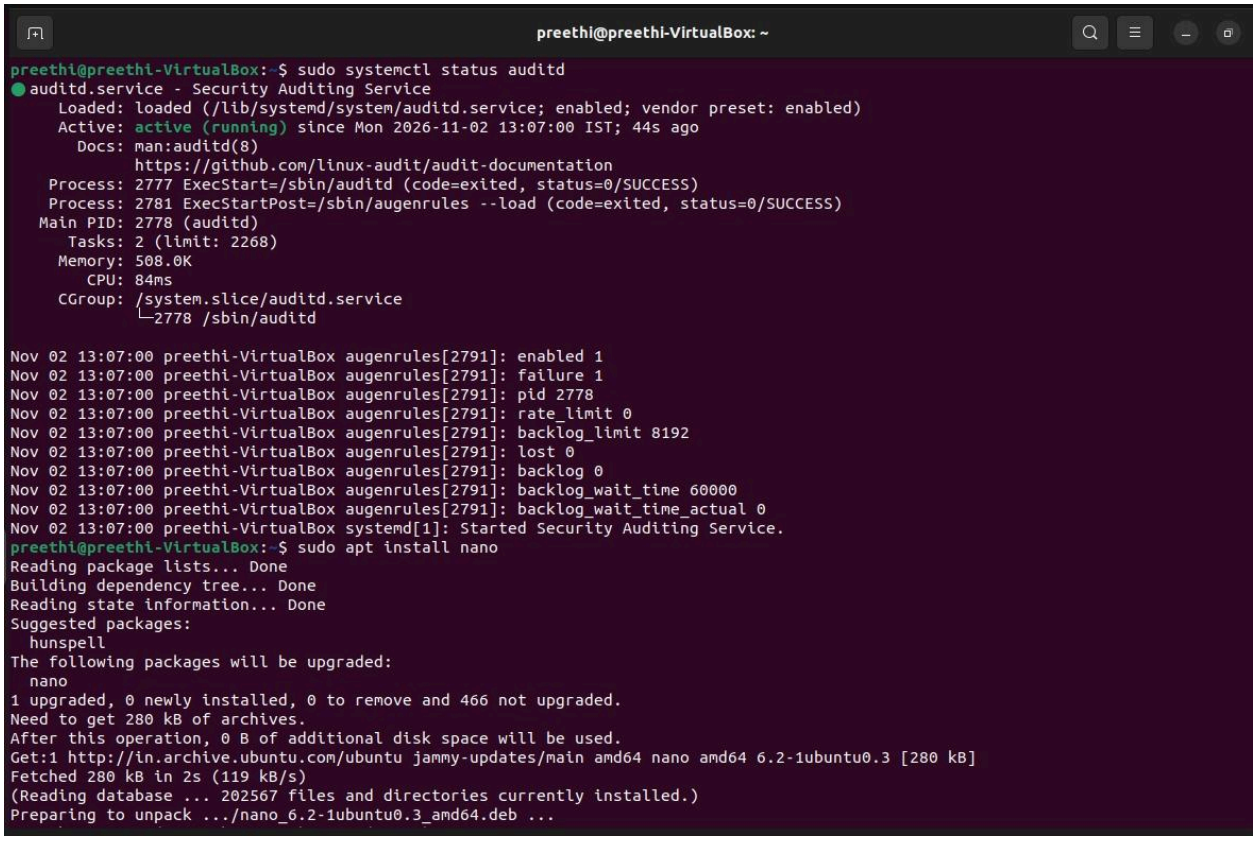
> Note: `ausearch -k nano` only returns results if an audit rule with the key `nano` exists. If no rule was added, use the message-type search `sudo ausearch -m EXECVE -ts recent`, or add a rule first: `sudo auditctl -a always,exit -F arch=b64 -S execve -k nano` and run the command again.

Screenshots

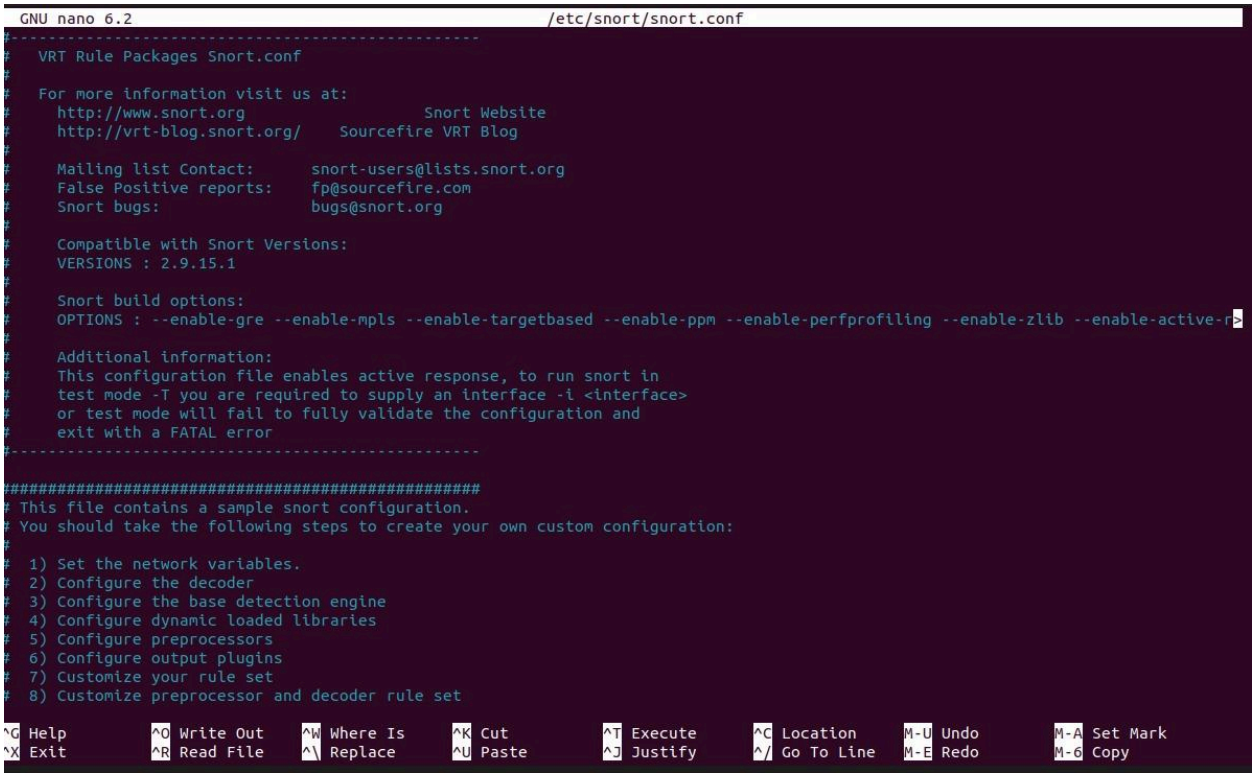
Screenshot 3.1 - auditd installation



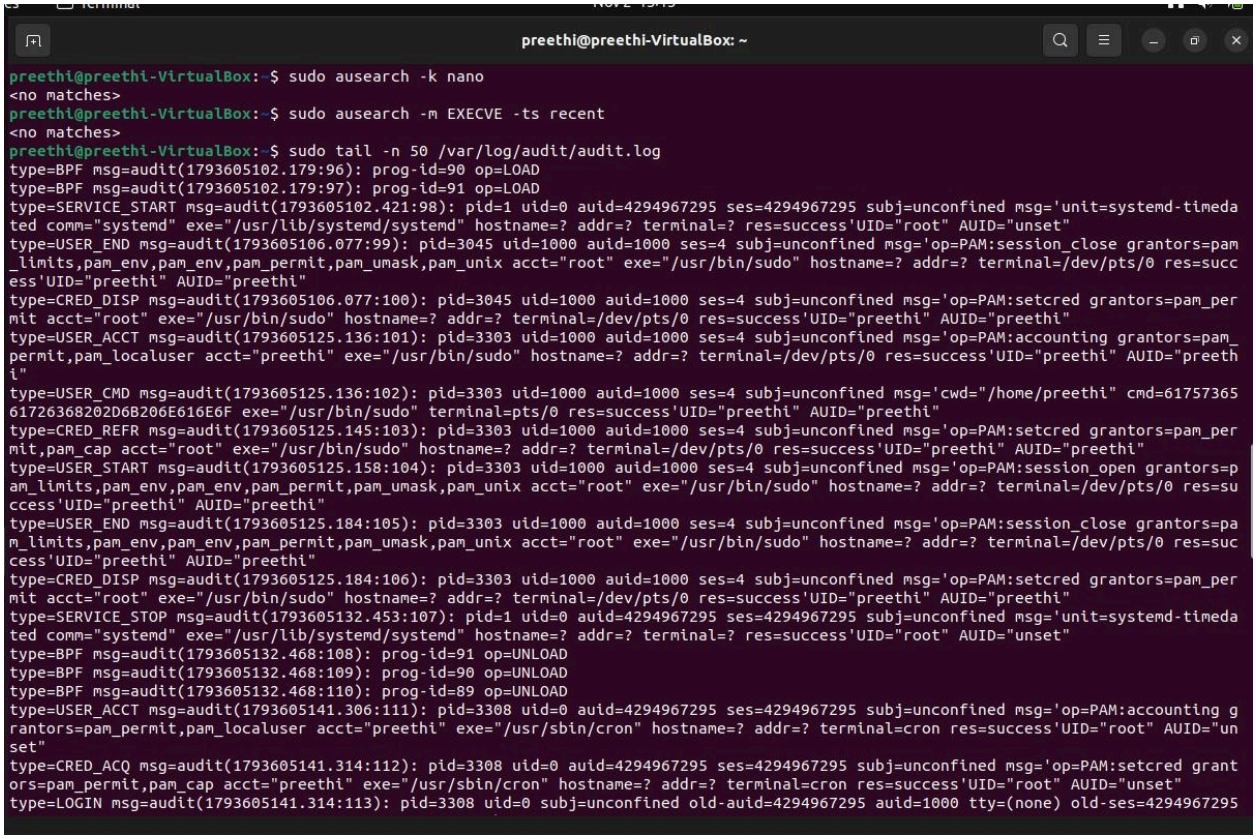
Screenshot 3.2 - `systemctl status auditd` showing active (running)



Screenshot 3.3 - Running `sudo apt install nano`



Screenshot 3.4 - `ausearch` output showing the command in the audit log



Audit log fields and why each is useful

Field	Meaning	Why it is useful
`type`	Kind of record (SYSCALL, EXECVE, PATH, CWD, USER_CMD, USER_AUTH, USER_LOGIN...)	Identifies what category of event this is
`msg=audit(epoch:serial)`	Timestamp (epoch) and unique event serial	Orders events and groups all lines of one event
`arch`	System architecture	Needed to interpret syscall numbers
`syscall`	System call number (e.g. 59 = execve on x86_64)	Shows the type of kernel action
`success` / `res`	Whether the action succeeded	Separates failed attempts from successful actions

Field	Meaning	Why it is useful
`exit`	Return code	Helps understand errors (e.g. permission denied)
`a0, a1, a2 ...`	Syscall arguments; in EXECVE records, the command and its arguments	Shows the full command line that was run
`argc`	Number of arguments	Confirms the command was fully captured
`pid`	Process ID	Identifies the process
`ppid`	Parent process ID	Builds the process tree
`auid`	Audit (login) user ID - the original logged-in user	Shows who really logged in, even after `sudo` or `su`
`uid` / `euid`	Real and effective user ID	Shows privilege level used (0 = root)
`gid` / `egid`	Real and effective group ID	Shows group privileges
`ses`	Session ID	Links all activity to one login session
`tty`	Terminal	Shows where the command was typed (pts/0, ssh)
`comm`	Short command name	Quick identification of the program
`exe`	Full path of the executable	Detects renamed or misplaced binaries
`key`	Rule key given with `-k`	Fast searching of events from a specific rule
`cwd` / `name` (PATH record)	Working directory and file path involved	Shows where and on which file the action happened
`hostname`, `addr`	Remote host and address (in login records)	Source of SSH logins

Findings

- auditd recorded the `apt install nano` command with the user (`uid` / `auid`), the executable (`exe`), the terminal (`tty`) and the arguments (EXECVE `a0`, `a1`, `a2`...).
- `auid` remained the original login user even though the command was run with `sudo`, which proves who performed the action.
- Multiple records (SYSCALL, EXECVE, CWD, PATH, PROCTITLE) share the same serial number and together describe one event.

SOC relevance

auditd provides kernel-level, tamper-resistant proof of who ran which command. Analysts use it to track privilege escalation, suspicious tool installation (`nmap`, `nc`, `nano` on sensitive systems), file tampering and login activity, and they forward it to a SIEM for correlation.